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StreamCatTools: An R package for working with StreamCat and LakeCat watershed data in R

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Summary

`StreamCatTools` provides functions for easily working with, visualizing and analyzing StreamCat (Hill et al., 2016) and LakeCat (Hill et al., 2018) watershed metrics within **R**. The StreamCat and LakeCat datasets provide hundreds of landscape metrics for both the local catchment (e.g. landscape draining to a particular stream reach) and full watershed (i.e., the local catchment and all upstream catchments) for every stream reach and lake depicted in the medium resolution National Hydrography Dataset Plus Version 2.1 (NHDPlusV21)(McKay et al., 2012) for the contiguous United States (CONUS). `StreamCatTools` functions wrap the application programming interface (API) for the StreamCat and LakeCat online database and facilitate their simple, straightforward access and use within R.

Statement of Need

Easily accessible, robust, and consistent watershed data underpin hydrology research, water quality monitoring programs, and predictive modelling applications, to name just a few examples. The StreamCat (Hill et al., 2016) and LakeCat (Hill et al., 2018) datasets fill this need by providing nationally consistent curated watershed data for CONUS that have had stringent quality control applied. The data encompass hundreds of watershed metrics for every stream reach and lake feature represented in the NHDPlusV21 (McKay et al., 2012). StreamCatTools provides easily accessible watershed metrics for CONUS via: (1) a simple interface to the StreamCat and LakeCat web services in the R programming language; (2) convenient functionality to find available StreamCat and LakeCat metric names and information about variables; and (3) extraction of StreamCat and LakeCat metrics by COMID (a unique identifier in the NHDPlusV21 framework), state, county, NHD hydro-region, or for all of CONUS. Providing these valuable watershed data via web services in R follows the FAIR principles (Findable, Accessible, Interoperable, and Reusable) laid out in (Wilkinson et al., 2016).

State of the Field

Programmatic access to watershed data in R in recent years had evolved to encompass several facets to *what* a package delivers: the hydrographic fabric (the stream reaches and catchments composing the network itself), observational time series (flow and water quality), general-purpose geospatial layers, and pre-computed landscape characteristics. The first role is well served by `hydrogeofetch` and its network-navigation companion `hydroloom` (Blodgett, 2026), which subset and traverse the NHDPlusV21 network. The second is served by `dataRetrieval` (DeCicco et al., 2026), which retrieves USGS and EPA hydrology and water-quality observations. The third is covered by general fetchers such as `FedData` and `elevatr`, which return raw federal layers (e.g., land cover, elevation, soils) that a user must summarize to catchments themselves.

The remaining role — landscape and anthropogenic metrics *already accumulated* to the local catchment and the full upstream watershed for every NHDPlusV21 or lake reach — has been a gap in the field, yet it is a form most analyses actually need. The StreamCat and LakeCat datasets fill this gap at a national scale, providing hundreds of natural and anthropogenic metrics for all NHDPlusV21 stream reaches, catchments, and waterbody basins in the conterminous United States. `StreamCatTools` is the primary programmatic entry point to these data. It wraps the StreamCat and LakeCat APIs so that researchers can retrieve analysis-ready watershed covariates in R without performing any flow-network accumulation themselves, and without manually assembling large geospatial layers. The package is used in water-quality modeling, biological condition assessment, and Clean Water Act applications.

Package Overview

`StreamCatTools` provides a simple streamlined set of functions to easily query and ingest watershed landscape metrics into an R session. Figure 1 shows the overall framework of the StreamCat database, application programming interface, and functionality in the package that simplifies data access in R using web services for the StreamCat and LakeCat datasets.

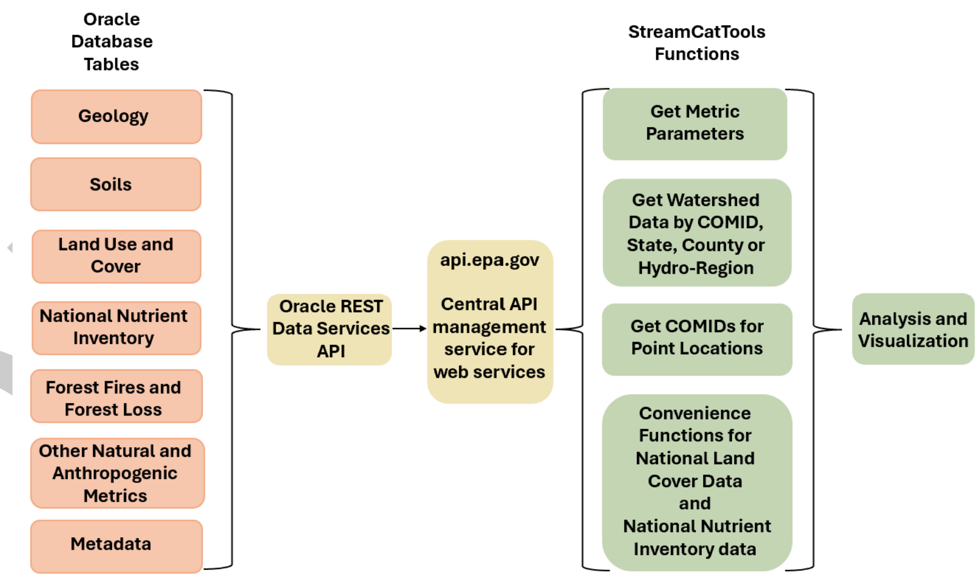

Figure 1: Diagram of the StreamCat and StreamCatTools framework. The backend Oracle database and REST web service are exposed through api.epa.gov to functions in StreamCatTools that simplify access and analysis of the data in R via the application programming interface (API). The example here shows the general categories of Oracle database tables in the StreamCat Oracle database.

The core functions in `StreamCatTools` leverage the `httr2` library (Wickham, 2026) for a modern, pipeable method for working with APIs. Specifically, `StreamCatTools` simplifies calls to the API for StreamCat and LakeCat data within the R programming language to allow a user to: (1) access details on available StreamCat and LakeCat metrics; (2) match users' sites to NHDPlusV21 waterbodies (streams or lakes) to access site-relevant StreamCat and LakeCat data; and (3) retrieve StreamCat and LakeCat watershed metrics by COMID, state, county, hydroregion, or for all of CONUS. Additionally, several convenience functions retrieve year-specific data for variables with time series data, such as the National Land Cover Database (NLCD) (U.S. Geological Survey, 2019) and the National Nutrient Inventory (NNI) (Brehob et al., 2025). Lastly, there is plotting functionality for use with NNI metrics as well as a function to retrieve the watersheds for specific lake features as an `sf` object. Users also can access watersheds for specific COMIDs using the `hydrogeofetch` package for visualization and plotting of watersheds along with StreamCat or LakeCat metrics.

You can install the most recent version of `StreamCatTools` from *CRAN* by running:

```
install.packages("StreamCatTools")
```

You can install the most recent package version from *GitHub* by running the following code (note we use `pak` as the replacement for `remotes` for package installation from GitHub):

```
install.packages("pak")
library(pak)
pkg_install("github::USEPA/StreamCatTools")
```

`StreamCatTools` is loaded into an **R** session:

```
library(StreamCatTools)
```

`StreamCatTools` includes several functions to retrieve basic metric metadata. First, users can list metric names and find out more about StreamCat and LakeCat data available in StreamCat using using the `sc_get_params()` and `lc_get_params()` functions. The parameters `aoi` and `metric_names` provide a user with the available areas of interest for metrics and names of metrics. The areas of interest available for all metrics in StreamCat and LakeCat are 'cat' and 'ws' (catchment and watershed). Some metrics in StreamCat are also available at the 100 meter riparian buffer scale ('catrp100' and 'wsrp100'), and for certain metrics the designation of 'other' needs to be used (where the metric is in-stream or derived and not part of the other listed areas of interest). Extracting this information in `StreamCatTools` looks like this:

```
aois <- sc_get_params(param='aoi')
aois
```

```
#> [1] "cat"      "catrp100" "other"    "ws"       "wsrp100"
```

```
names <- sc_get_params(param='metric_names')
names[1:10]
```

```
#>  [1] "agkffact"      "al2o3"         "bankfulldepth" "bankfullwidth"
#>  [5] "bfi"           "canaldens"     "cao"           "cbnf"
#>  [9] "chem"          "clay"
```

And the same information for LakeCat metrics is derived in similar fashion:

```
aois <- lc_get_params(param='aoi')
aois
```

```
#> [1] "cat" "ws"
```

```
names <- lc_get_params(param='metric_names')
names[1:10]
```

```
#>  [1] "agkffact"     "al2o3"        "bfi"          "canaldens"    "cao"
#>  [6] "cbnf"         "clay"         "coalminedens" "compstrgth"   "damdens"
```

StreamCat and LakeCat are built around the concepts of local drainage area (i.e catchment) and watershed (i.e., the local drainage area and all upstream catchments) (Hill et al., 2016). This approach uses the NHDPlusV21 hydrographic framework of catchments as the building block for summarizing information represented in landscape data. Watershed data are produced by combining both the catchment landscape summary and all upstream catchment summaries using either the weighted average of metrics (in most cases) or a sum or count (for certain metrics). `sc_get_params()` and `lc_get_params()` also include a 'variable_info' parameter to return more detailed metadata for metrics including both short and long metric descriptions, years available (if applicable), units, and the metric category (Table 1).

Table 1: Example of variable information returned by the variable_info parameter in the sc_get_params function.

Metric	Short Description	Year	Units
agkffact[AOI]	Ag Soil Erodibility Kf Factor	NA	Unitless
bfi[AOI]	Base Flow Index	NA	Percent
huden[Year][AOI]	Mean Housing Density	2010	Count/Square Kilometer
inorgnwetdep[Year][AOI]	Mean Annual Precipitation-Weighted Nitrogen	2008	Kilogram/Hectare/Year
n_ags_[Year][AOI]	Nitrogen Agricultural Surplus	1987-2017	Kilograms
pcthighsev[Year][AOI]	Percent High Burn Severity Class For Year	1984-2018	Percent

Additional functions for getting metadata about the underlying StreamCat and LakeCat data include the `sc_fullname()` and `lc_fullname()` functions which provide the descriptive full name for any given metric in StreamCat or LakeCat:

```
sc_fullname(metric='pctgrs2019')
```

```
#> [1] "Grassland/Herbaceous Percentage 2019"
```

Users can also filter metric names and information by the metric year(s), the indicator categories for metrics, the metric dataset names, or the areas of interest available for a given metric using the `sc_get_metric_names` or `lc_get_metric_names` functions (Table 2).

```
metrics <- sc_get_metric_names(category = c('Deposition','Climate'),
                               aoi=c('Cat','Ws'))
my_data <- head(metrics[,c('Category','Metric','AOI')],10)
```

Table 2: Example of metric names returned by the sc_get_metric_names function.

Category	Metric	AOI
Climate	bfi[AOI]	Cat, Ws
Deposition	inorgnwetdep[Year][AOI]	Cat, Ws
Deposition	nh4[Year][AOI]	Cat, Ws
Deposition	no3[Year][AOI]	Cat, Ws
Climate	precip8110[AOI]	Cat, Ws
Climate	precip9120[AOI]	Cat, Ws
Climate	precip[Year][AOI]	Cat, Ws
Deposition	sn[Year][AOI]	Cat, Ws
Climate	tmax8110[AOI]	Cat, Ws
Climate	tmax9120[AOI]	Cat, Ws

More details on these functions can be found at the package introduction page.

The primary package functionality is in the `sc_get_data` and `lc_get_data` functions which allow users to extract catchment or watershed metrics of interest by providing NHDPlusV21 COMIDs for streams or lakes within the database. The following example shows a request for two catchment and watershed metrics for three stream reaches:

```r
df <- sc_get_data(metric='pcturbmd2019,damdens',
                  aoi='cat,ws',
                  comid='179,1337,1337420')
```

The first line requests percent of NLCD medium intensity developed land cover in 2019 and the density of dams. The second line specifies that the geographic area of interest (aoi) is the drainage to the local reach scale (i.e. 'cat', short for catchment) and the full watershed. The final line requests these data for three NHDPlusV21 stream segments specified by their unique COMIDs. A similar request for LakeCat metrics for data on lakes might look like this, where a user can ask for data for a particular county using the county FIPS code rather than for a set of identifying COMIDs:

```r
df <- lc_get_data(metric='pctwdwet2006', aoi='ws', county='41003')
head(df)
```

```
#>      comid pctwdwet2006ws
#> 1 23769033      0.0000000
#> 2 23769029      0.1701645
#> 3 23769021      2.4697951
#> 4 23769003      0.0000000
#> 5 23768999      8.1185071
#> 6 23768983      0.0000000
```

Users can also request data for one or more state(s), county(ies), hydroregion(s), or all of CONUS (though asking for data for all of CONUS may time out if asking for too many metrics at once). The following example demonstrates a request data for a particular county in Oregon:

```r
df <- sc_get_data(metric='pctwdwet2006', aoi='ws', county='41003')
```

StreamCatTools provides a convenience function in either `sc_get_params` or `lc_get_params` to discover the FIPS codes for states and counties to make it easier for users to query data by these parameters. For example, the following code chunk shows how to get the FIPS code for Benton County, Oregon to query data for that county:

```r
df <- sc_get_params(param='county') |> dplyr::filter(state == 'OR' & county_name == 'Ben
```

Or we can ask for several different metrics for all of a state:

```r
# State-wide watershed queries are large; run the live call once, then
# cache to disk so subsequent knits read the cache instead of
# re-hitting (or stalling on) the API. Commit the .rds alongside paper.Rmd.
ct_cache <- "clay_agkffact_CT_ws.rds"
if (file.exists(ct_cache)) {
  df <- readRDS(ct_cache)
} else {
  df <- sc_get_data(metric = 'clay,agkffact', aoi = 'ws', state = 'CT')
  saveRDS(df, ct_cache)
}
head(df)
```

```
#>     comid agkffactws   clayws
#> 1 7702768     0.0177 6.100716
#> 2 7702200     0.0009 6.243493
#> 3 7702202     0.0011 6.368938
#> 4 7701112     0.0177 6.080299
#> 5 7701168     0.0054 6.092434
#> 6 7701214     0.0040 6.082292
```

Additionally, convenience functions are provided for accessing the NLCD and NNI datasets, respectively, using `sc_get_nlcd` and `lc_get_nlcd` and `sc_get_nni` and `lc_get_nni`:

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```r
df <- sc_get_nlcd(comid='1337420', year='2019', aoi='ws')
```

```r
df <- sc_get_nni(year='1987, 1990, 2005, 2017', aoi='cat,ws',
                 comid='179,1337,1337420')
```

Software design

StreamCatTools is designed as a light convienence wrapper over the StreamCat and LakeCat REST services with all requests issued through httr2. The interface is organized into two symmetric families of functions: sc_* for stream/catchment data from StreamCat and lc_* for lake-basin data from LakeCat.

Data access is handled by sc_get_data() and lc_get_data(), which query by metric name, area of interest (catchment or watershed), and spatial filter (COMID, county, state, or hydroregion). These functions batch requests transparently so that hundreds of COMIDs can be passed in a single call without overrunning server limits, apply input validation with informative error messages, and return tidy data frames ready for analysis. Discovery and metadata are handled by sc_get_params() and lc_get_params(), which enumerate available metrics and their full descriptions and expose lookup tables for state and county names and FIPS codes, so that queries can be constructed reproducibly without hard-coding parameters.

Results are returned as sf objects, and the package interoperates with hydrogeofetch for network context and spatial integration. For lake basins, lc_get_watershed() retrieves watershed polygons from an S3-hosted, HUC2-partitioned GeoParquet store using DuckDB with multithreading, exponential backoff, and retries — a cloud-native path that avoids transmitting large geometries through the API.

Applications and Discussion

One can easily summarize the data returned from sc_get_data or lc_get_data for a set of rivers or lakes or for the watersheds for an entire county or state in a number of ways. The following example calculates the mean percent of agriculture on slopes > 10% within a watershed for all watersheds in Benton County, Oregon:

```r
df <- sc_get_data(metric='pctagslphigh2019', aoi='ws', county='41003') |>
  dplyr::summarise(mean_pctagslphigh2019ws = mean(pctagslphigh2019ws, na.rm=TRUE))
df
```

```
#>   mean_pctagslphigh2019ws
#> 1                4.406472
```

Watershed metrics from StreamCatTools can also be easily visualized with functions from hydrogeofetch (Blodgett, 2026) and ggplot2 (Wickham, 2016).

```r
library(hydrogeofetch)
library(ggplot2)
library(ggspatial)
library(StreamCatTools)

start_comid = 23763517
nldi_feature <- list(featureSource = "comid", featureID = start_comid)
flowline_nldi <- hydrogeofetch::navigate_nldi(nldi_feature, mode = "UT", data_source = "
df <- sc_get_data(metric='pctimp2019', aoi='cat', comid=flowline_nldi$UT_flowlines$nhdpl
flowline_nldi <- flowline_nldi$UT_flowlines
flowline_nldi$PCTIMP2019 <- df$pctimp2019cat[match(flowline_nldi$nhdplus_comid, df$comid
basin <- hydrogeofetch::get_nldi_basin(nldi_feature = nldi_feature)
```

Figure 2 plots the NLCD percent imperviousness (percentage of area covered by constructed, artificial surfaces) for the local drainage (catchment in NHDPlusV21 syntax) and displays the values mapped to each stream reach and to the overall basin boundary.

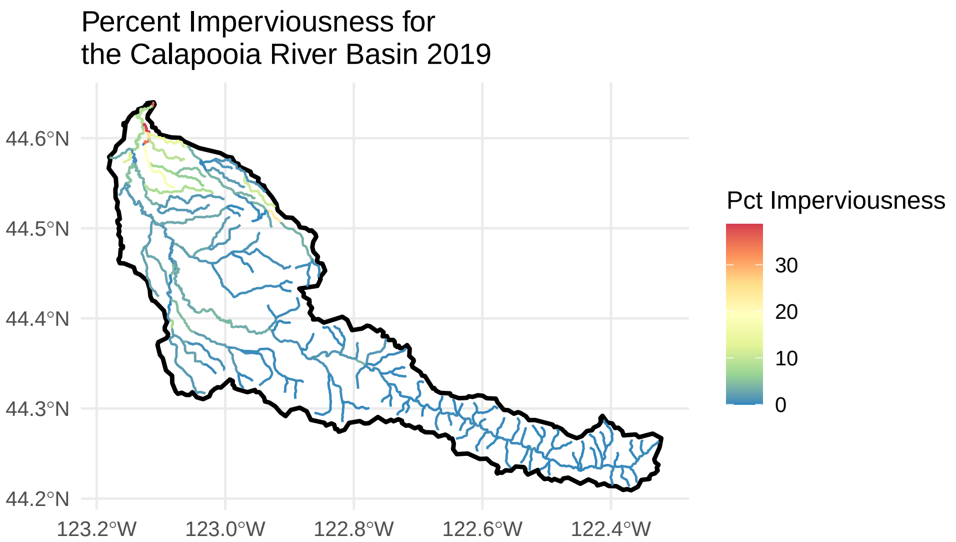

Figure 2: Map of NLCD percent imperviousness for each catchment for the Calapooia River watershed in Oregon.

Watersheds for lakes can also be retrieved using the `lc_get_watershed` function to visualize LakeCat metrics along with the plotted watersheds for lake features as shown in Figure 3.

```
library(ggplot2)
library(patchwork)
library(ggforce)

df <- lc_get_nlcd(comid='19334077', year='2019', aoi='ws')
lake <- hydrogeofetch::get_waterbodies(id = 19334077)
ws <- lc_get_watershed(comid = 19334077, huc2 = "01",huc2_filter = "01",
                       threads = 2,retries = 5, verbose=FALSE, progress=FALSE)
```

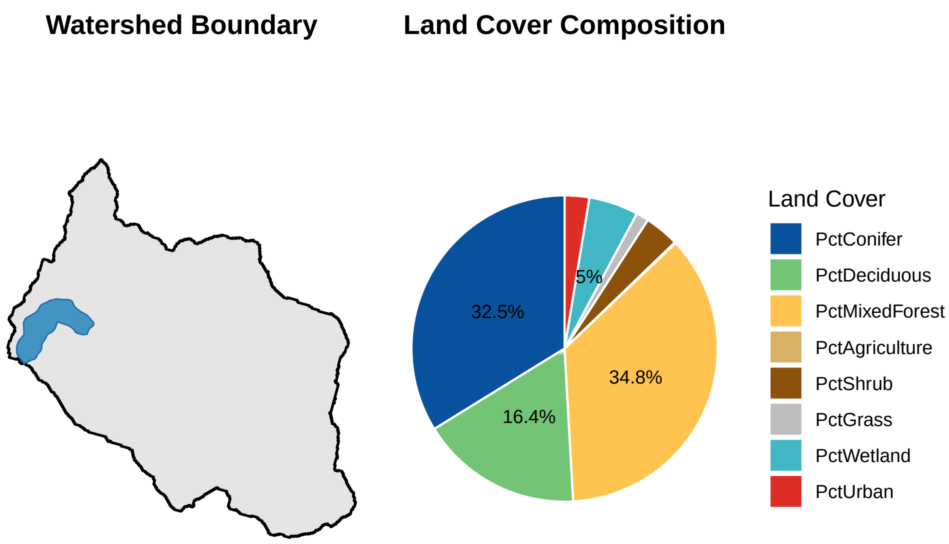

Figure 3: NLCD land cover proportions with an example lake watershed.

Functions to specifically plot NNI metrics in StreamCat have been developed and are also available (Markley et al., 2026), such as Figure 4 showing the annual time series of nitrogen and phosphorus budget data for a given watershed such as the Mississippi-Atchafalaya River Basin:

```
library(StreamCatTools)
library(ggplot2)
library(ggpattern)
com <- '22812041'
sc_plotnni(comid = com, include.nue = TRUE)
```

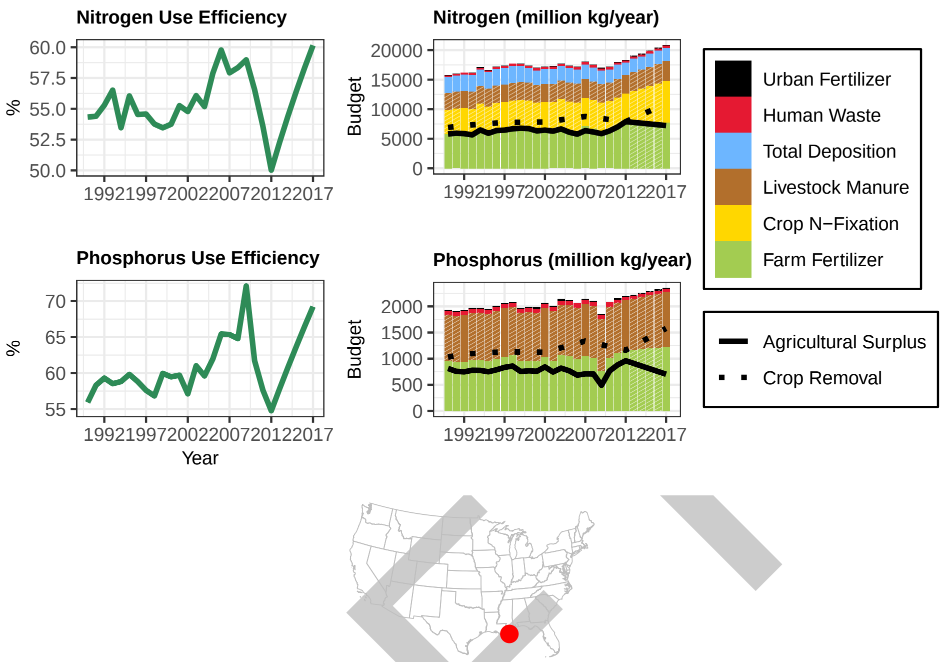

Figure 4: Annual time series of nitrogen and phosphorus budget data for the Mississippi-Atchafalaya River Basin.

Potential future functionality for `StreamCatTools` includes expanding the scope of plotting functions as well as expanding the range of metrics and ease of querying these metrics. Additionally, a complementary package, `StreamCatR` is being developed that will allow users to quickly and easily process their own landscape metrics to the NHDPlus framework or other hydrologic frameworks with network topology. `StreamCatTools` currently facilitates easy ingestion of StreamCat and LakeCat watershed landscape metrics into workflows in R which is of great use to state watershed planners, researchers and non-governmental organizations and borne out by the over 5000 package downloads and over 350 citations of the underlying StreamCat and LakeCat data that are served by the `StreamCatTools` package.

Research impact

`StreamCatTools` and the underlying StreamCat dataset provides more than 600 natural and anthropogenic landscape metrics — summarized to both local catchments and full upstream watersheds — for roughly 2.65 million stream reaches across the conterminous United States (Hill et al., 2016), with the parallel LakeCat dataset extending the same framework to more than 356,000 lakes. These data underpin national-scale products including models of biological stream condition, indices of catchment and watershed integrity, and predicted stream temperature, and they are widely used by researchers and resource managers precisely because they remove the specialized geospatial expertise otherwise required to accumulate landscape information along the stream network. Beyond research, the metrics support management and regulatory applications under Clean Water Act authorities, including antidegradation and healthy-watersheds assessments, as well as species-distribution and aquatic-connectivity studies.

`StreamCatTools` provides a primary programmatic pathway to these data - StreamCat and LakeCat are accessed through web tools, the REST API, and this R package. By exposing metric discovery, batched retrieval, and spatial integration with NHDPlusV21 in a single reproducible interface, the package lowers the barrier to using a heavily relied-upon national dataset and makes the workflows that feed condition assessments and management decisions transparent and repeatable.

Acknowledgements

Examples of using StreamCat and LakeCat make extensive use of `hydrogeofetch` (Blodgett, 2026) and the functions for accessing the API are facilitated through use of `httr2` (Wickham, 2026). Figures were created using `ggplot2` (Wickham, 2016).

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AI Usage Disclosure

The R code for this R package was initially developed by Marc Weber and minimal generative AI was used for the most recent version of the package where several functions were refactored.

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